

4-5.9

Ratio & Proportion

with (sub)

DATE:

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2 quantities will
have ratio

$$P : Q \Rightarrow \frac{P}{Q}$$

$$Q : P \Rightarrow \frac{Q}{P}$$

$$a : b = c : d$$

$$\frac{a}{b} = \frac{c}{d}$$

$$a : b :: c : d$$

$$y) \frac{a}{b} = \frac{c}{d}$$

(dividend)

$$\frac{a-b}{b} = \frac{c-d}{d}$$

$$\frac{a+b}{b} = \frac{c+d}{d}$$

$$\frac{a+b}{a-b} = \frac{c+d}{c-d}$$

In simple

Compound → Dividend

$$\frac{a}{b} = \frac{c}{d} \Rightarrow \frac{a+b}{a-b} = \frac{c+d}{c-d}$$

~~Tip~~

1) If they are proportional

$$\frac{a}{b} = \frac{c}{d} \Rightarrow ad = bc$$

$$2) \frac{a}{b} < \frac{c}{d}$$

$$\text{if } ad > bc \Rightarrow \frac{a}{b} > \frac{c}{d}$$

$$\text{if } ad < bc \Rightarrow \frac{a}{b} < \frac{c}{d}$$

$$\text{if } ad = bc \Rightarrow \frac{a}{b} = \frac{c}{d}$$

Inverted

$$\frac{a}{b} = \frac{c}{d} \Rightarrow \frac{b}{a} = \frac{d}{c}$$

3) ratio of 3 Nos

$$a : b : c \rightarrow a : b :: b : c$$

1 2 3 (Extreme)
non

4) 4 Nos

$$a : b : c : d$$

1 2 3 4 — Extreme
mean

$$3) \frac{a}{b} = \frac{c}{d} \text{ [Compound rule]}$$

$$\frac{a+b}{b} = \frac{c+d}{d}$$